

# M2C2kit: A Flexible, Cross-Platform Engine for Cognitive Assessment

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## Software

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## Summary

Accurate and efficient measurement of brain health is critical for advancing research and clinical practice across multiple disciplines, including cognitive aging, neuropsychology, neuroscience, and medicine.

The m2c2kit project ([M2C2 Project, n.d.b](#)) is an open-source framework, released under the Apache 2.0 license, for developing and deploying mobile cognitive and motor assessments ([M2C2 Project, n.d.a](#)). Assessments can be delivered through modern web browsers and integrated into common research data-collection platforms, including Qualtrics and REDCap, or embedded within WebViews in native mobile applications.

m2c2kit is designed to make cognitive and motor assessments portable across smartphones, tablets, and computers while preserving consistent appearance and behavior. Assessments developed using m2c2kit and prior iterations of the framework have been used in published research on ambulatory and mobile cognitive assessment (e.g., ([Sliwinski et al., 2018](#)), ([Cerino et al., 2021](#)), ([Thompson et al., 2022](#)), ([Hakun et al., 2023](#)), ([Shaleha & Roque, 2024](#))).

The framework uses Google's CanvasKit WebAssembly package ([Canvaskit-Wasm, n.d.](#)), based on the Skia 2D graphics engine, to render assessments within an HTML canvas. This architecture supports consistent rendering across devices and operating systems as well as graphical, animated, and interactive assessment paradigms. m2c2kit also provides documentation, example assessments, tutorials, and a command-line interface for scaffolding assessments and generating deployable web bundles.

## Statement of Need

m2c2kit addresses the need for flexible and scalable tools for cognitive and motor assessment within ambulatory research protocols. Ambulatory cognitive assessment involves administering brief cognitive tasks repeatedly throughout participants' daily lives. These intensive measurements allow researchers to characterize short-term fluctuations and patterns in cognition that may be associated with disease processes, cognitive resilience, or contextual factors such as stress, sleep, and physical activity. Compared with assessment restricted to laboratory settings, this approach can increase ecological validity and provide measurements at times and places relevant to participants' everyday functioning.

Despite the potential of mobile cognitive assessment, implementing these protocols at scale remains challenging. Existing approaches may require specialized hardware, substantial licensing or development costs, or systems that are difficult to adapt to the requirements of a particular

41 research study. m2c2kit provides a modular framework in which developers can create and  
42 modify assessments using TypeScript/JavaScript while retaining control over assessment  
43 behavior, appearance, timing, and data generation.

44 The intended users include researchers, research software developers, and clinicians who need  
45 customizable cognitive or motor assessments that can operate across heterogeneous participant  
46 devices. Assessments can run directly in modern web browsers, as components of protocols  
47 incorporating platforms such as REDCap or Qualtrics, or inside WebViews controlled by native  
48 mobile applications.

## 49 State of the Field

50 Several existing tools support computerized behavioral and cognitive experiments. Open-source  
51 libraries such as jsPsych ([Leeuw & colleagues, n.d.](#)) and lab.js ([Henninger & colleagues, n.d.](#)),  
52 as well as systems such as Gorilla ([Gorilla Task Builder, n.d.](#)), provide important capabilities  
53 for creating and administering browser-based experiments. These tools have substantially  
54 expanded access to online behavioral research.

55 m2c2kit addresses a complementary design problem: delivering highly interactive and graphics-  
56 intensive cognitive assessments from a common code base across browsers and native mobile  
57 WebViews. Rather than constructing an assessment primarily from standard HTML interface  
58 elements, m2c2kit renders assessment scenes through a graphics engine. This permits precise  
59 control over stimulus appearance, animation, interactions, and physics while reducing platform-  
60 dependent differences in the presentation of HTML elements.

61 The decision to develop a separate framework rather than implement these capabilities as  
62 extensions to an existing experiment library reflects this architectural distinction. m2c2kit  
63 requires a scene-oriented rendering and interaction model in which drawing, animation, frame  
64 timing, input, and assessment events share a common abstraction. The framework borrows  
65 patterns and naming conventions from Apple's SpriteKit but is not a port of SpriteKit. Like  
66 SpriteKit, m2c2kit abstracts drawing and frame-timing operations so developers can focus on  
67 assessment logic and experimental paradigms.

68 This architecture makes m2c2kit particularly appropriate for assessments involving dynamic  
69 stimuli, animations, or interactions for which rendering consistency is an important measurement  
70 consideration. It also permits the same assessment package to be used in browser-based studies  
71 and as the assessment component of a native application.

## 72 Software Design

73 m2c2kit is organized around a common TypeScript code base that is compiled into JavaScript  
74 packages and distributed through npm ([Npm, n.d.](#)). Researchers and developers install these  
75 packages to construct assessments, and the resulting code can be bundled into a static web  
76 application for browser delivery or loaded within a WebView in a native application. Figure 1  
77 summarizes these deployment paths.

78 A central design decision is the use of CanvasKit ([Canvaskit-Wasm, n.d.](#)) and its Skia-based  
79 graphics engine. Rendering assessment stimuli to a canvas rather than relying primarily on  
80 platform-specific HTML controls provides a common graphical environment across desktop  
81 and mobile browsers. The trade-off is that m2c2kit maintains its own scene and interaction  
82 abstractions rather than delegating presentation to the browser's document interface. This  
83 additional abstraction is useful for research applications in which control over stimulus  
84 presentation, animation, and interaction is important.

85 A second design decision is to separate assessment logic from the environment responsible for  
86 deployment and participant-data management. In a browser-only workflow, the assessment

87 can post participant data to a researcher's web service. In a native application, the WebView  
88 executes the same assessment while native code can manage lifecycle events, local storage,  
89 participant information, and synchronization with external services. This permits researchers  
90 to reuse an assessment across deployment environments without maintaining separate browser,  
91 iOS, and Android implementations.

92 m2c2kit also emphasizes reproducible software distribution. Core logic is covered by unit tests,  
93 while integration tests use browser automation to verify functionality, rendering, and behavior.  
94 Package releases are published to npm through a GitHub Actions CI/CD pipeline rather than  
95 through manual deployment. The immutable versioned packages distributed by npm therefore  
96 provide a common executable source for assessments running in mobile browsers, desktop  
97 browsers, or embedded WebViews.

98 The project additionally provides a command-line interface for scaffolding assessments and  
99 generating deployments, along with documentation covering graphical primitives, assessment  
100 development, and integration with research platforms. These components are intended to  
101 reduce the amount of platform-specific infrastructure that individual research teams must  
102 implement and maintain.

## 103 Research Impact Statement

104 m2c2kit and earlier iterations of its assessment framework have supported published research  
105 using ambulatory and mobile cognitive assessment. Applications include studies of ambulatory  
106 cognitive variability (Sliwinski et al., 2018), remote cognitive assessment (Thompson et al.,  
107 2022), working-memory measurement (Hakun et al., 2023), and other research using repeated  
108 digital cognitive measurements (Cerino et al., 2021; Shaleha & Roque, 2024). Earlier versions  
109 of the framework have also been used in research evaluating device-related measurement  
110 characteristics and latency (Nicosia et al., 2023).

111 The framework has been adopted across research projects involving investigators at different  
112 career stages and technical skill levels. Its deployment model supports integration into  
113 established research workflows, including REDCap and Qualtrics, while its WebView interface  
114 permits integration into native data-collection applications. MetricWire has used this approach  
115 to embed assessments within its mobile application environment.

116 These uses demonstrate that m2c2kit functions as reusable research infrastructure rather than  
117 software developed for a single analysis or study. Its common rendering architecture and  
118 package-based distribution allow assessment implementations to be reused across projects  
119 and participant platforms, while its open-source design allows researchers to inspect, modify,  
120 and extend the software for new assessment paradigms. Additional information about the  
121 framework and its flexibility is available in the project whitepaper (Unknown, n.d.).

## 122 Figures

123 Figure 1 depicts the m2c2kit software distribution and deployment architecture. Source  
124 TypeScript is compiled and distributed as JavaScript packages through npm. Researchers can  
125 use these packages to build browser-based assessments or integrate the same assessment code  
126 into native applications through a WebView. In both deployment modes, participant data can  
127 be transmitted to researcher-controlled web services.

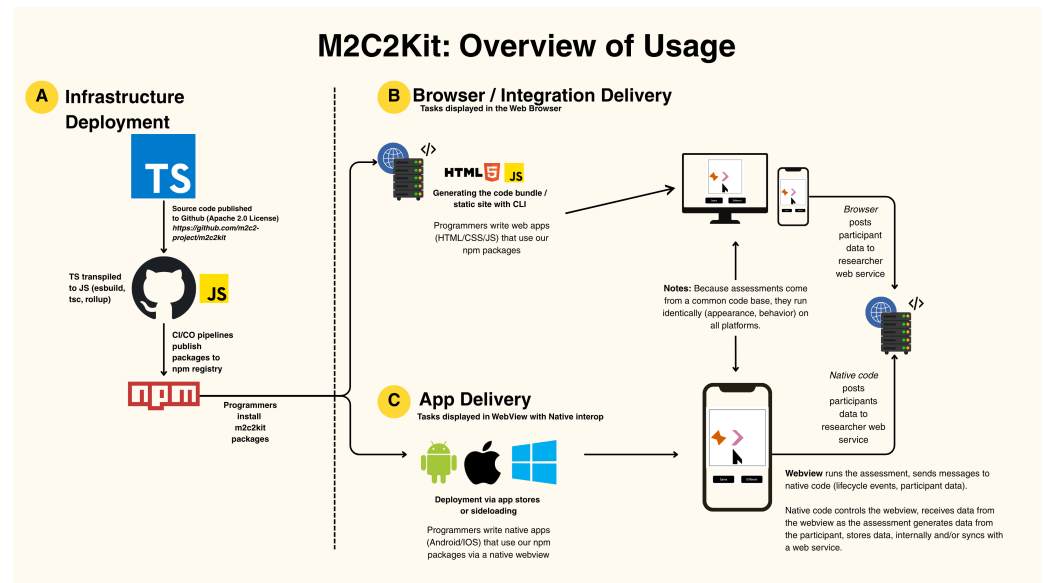


Figure 1: Overview of m2c2kit software distribution and browser and native-app deployment workflows.

## AI Usage Disclosure

Generative AI tools were used for code review and copy-editing portions of this manuscript. All AI-assisted code and text was reviewed and edited by the authors. The authors are responsible for the final content of the manuscript and made the core scientific, architectural, and software design decisions described here.

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